

GOVERNMENT ENGINEERING COLLEGE, BHARUCH

Student's Weekly Report of Internship

Student Name:	PATEL BINIT UMESHKUMAR
Enrollment Number:	230143107014
Week Duration:	January 8, 2026 - January 14, 2026 (Week 2)
Name Of Organization:	Microsoft Elevate Program (via Edunet Foundation & FICE Education)
External Guide Name:	Adarsh Gupta
External Guide Contact:	0124-4080107

Work done in last Week with description:

1. Week 2 Module: Data Cleaning & Preprocessing

Description:

Attended live session on "Data Cleaning & Preprocessing" which covered techniques for handling real-world messy datasets. Learned about missing value imputation, outlier detection, data normalization, and feature engineering.

Key Topics Covered:

- Identifying and handling missing data (dropna, fillna strategies)
- Outlier detection using IQR and Z-score methods
- Data normalization techniques (Min-Max scaling, Standardization)
- Handling categorical data (One-Hot Encoding, Label Encoding)
- Date-time data processing

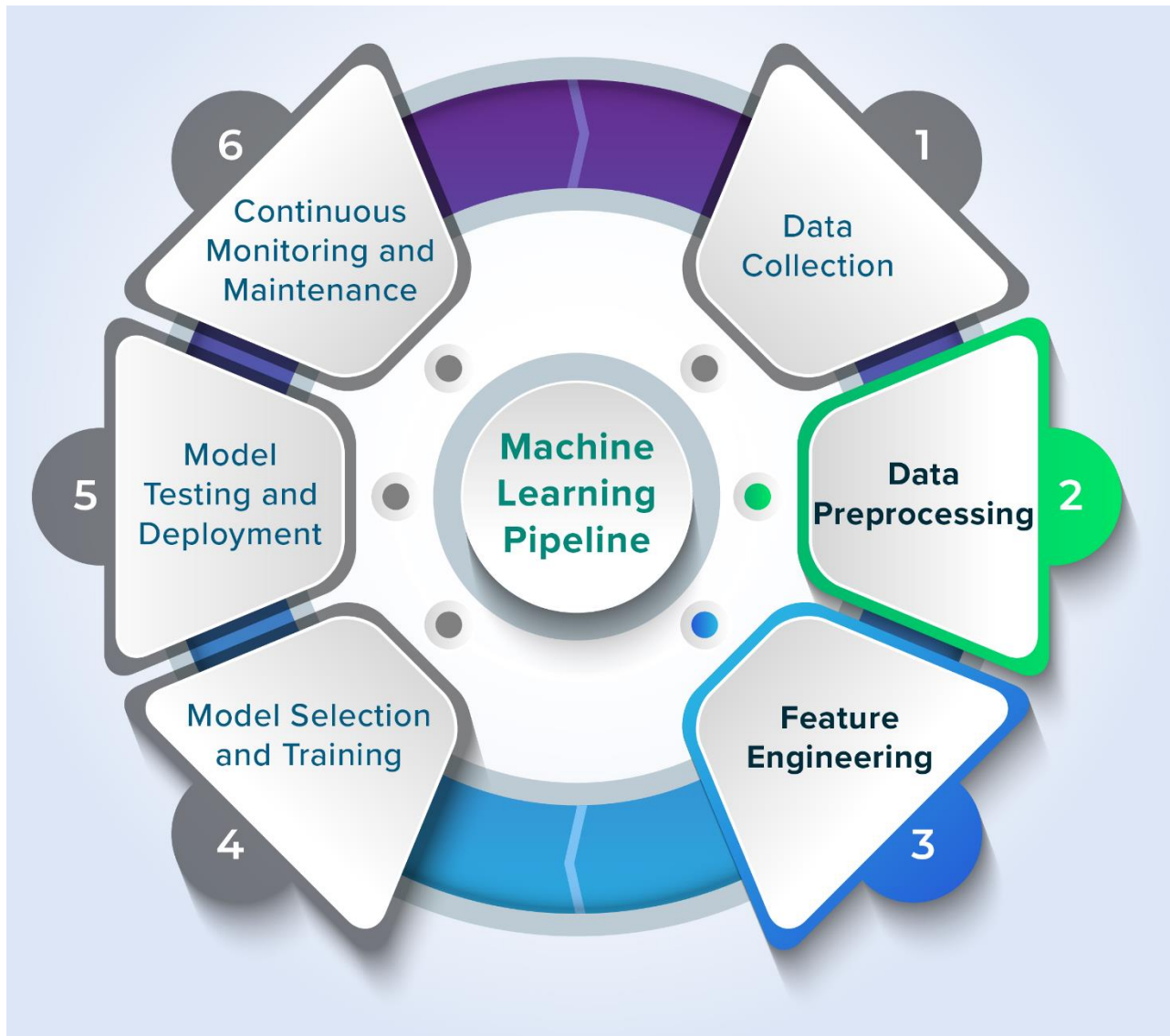


Figure 1: Data Preprocessing Pipeline

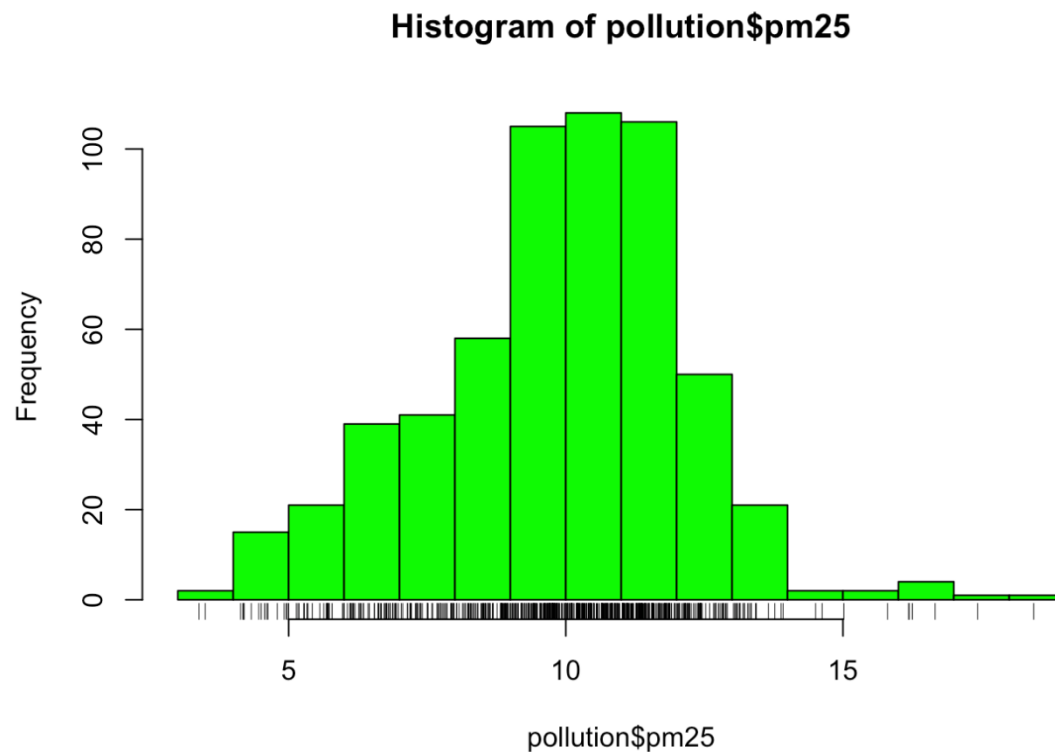
2. Week 2 Module: Exploratory Data Analysis (EDA)

Description:

Participated in "Exploratory Data Analysis (EDA)" session focusing on understanding data distributions, relationships, and patterns through statistical analysis and visualization.

Key Topics Covered:

- Statistical summary (mean, median, std, quantiles)
- Data distribution analysis (histograms, box plots)
- Correlation analysis and heatmaps
- Identifying patterns and trends in data
- EDA best practices and reporting



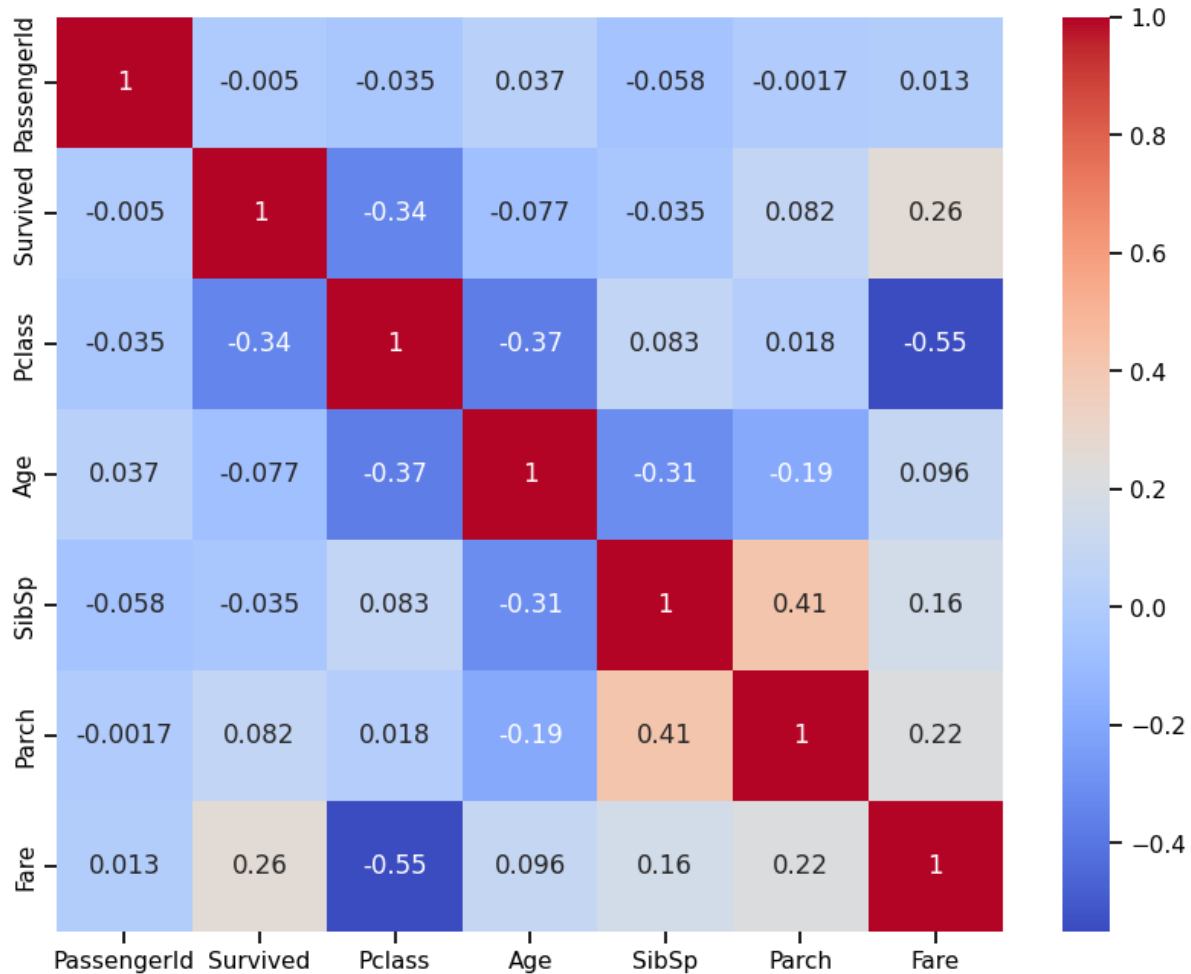
3. Week 2 Module: Data Visualization with Matplotlib & Seaborn

Description:

Completed "Data Visualization" module learning to create effective visualizations for data communication using Python libraries Matplotlib and Seaborn.

Key Visualizations Learned:

- Line plots for time-series data
- Bar charts for categorical comparisons
- Scatter plots for relationship analysis
- Heatmaps for correlation matrices
- Box plots for outlier visualization
- Customizing plots (titles, labels, colors, legends)



4. GTU Project Ideation and Planning (Border Defence Domain)

Description:

Finalized project idea for GTU internship under "Border Defence and Surveillance" problem domain. Conducted detailed planning session and created project roadmap.

Project Title:

"AI-Powered Border Surveillance System with Real-Time Anomaly Detection and Cloud-Based Alert Management"

Work Completed:

- Analyzed GTU problem statement requirements thoroughly
- Researched existing AI-based surveillance solutions
- Identified suitable public datasets (UCF-Crime, xView, DOTA)
- Defined project objectives using YOLOv8 and Ensemble ML
- Planned Azure Cloud deployment and Power BI dashboard
- Created 12-week implementation timeline
- Drafted project proposal outline

AI-Powered Border Surveillance System Architecture

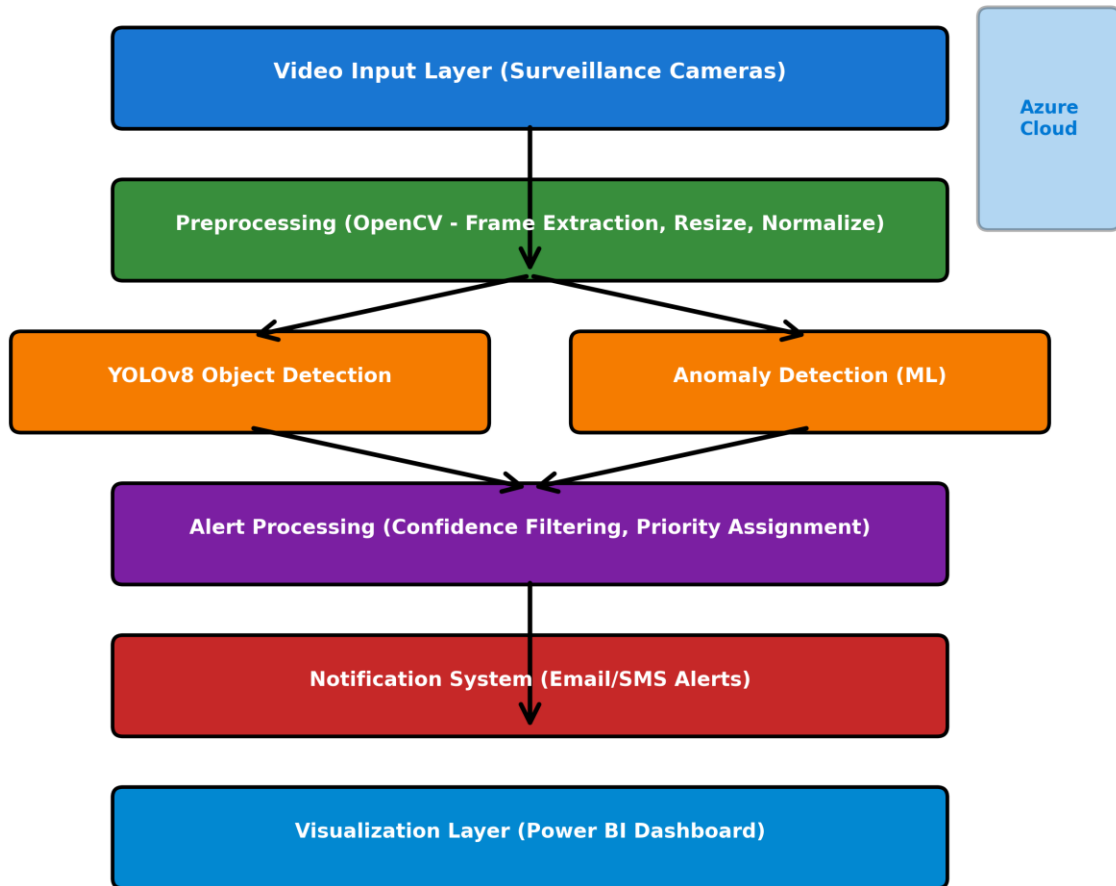


Figure 2: AI-Powered Border Surveillance System Architecture

5. Research on Datasets and Technologies

Description:

Conducted research on available datasets for border surveillance and object detection, as well as relevant technologies and frameworks.

Datasets Explored:

- **UCF-Crime Dataset:** 1,900 videos with 13 anomaly types for training anomaly detection models
- **xView Satellite Dataset:** 1 million+ annotated objects in 0.3m resolution satellite imagery
- **DOTA Dataset:** 280,000 annotated objects in 15 categories for aerial detection

Technologies Researched:

- **YOLOv8:** Latest object detection architecture with 30+ FPS processing

- **Azure Functions:** Serverless computing for video processing
- **Azure Blob Storage:** Scalable video/image storage solution
- **Power BI:** Interactive dashboard for real-time alerts and analytics

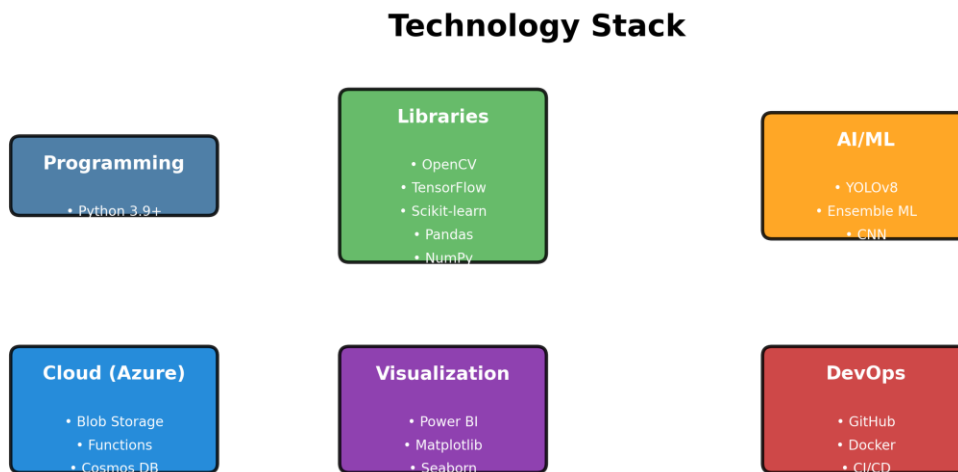


Figure 3: Complete Technology Stack for the Project

6. Internship Presentation Preparation

Description:

Started preparing presentation for 2-week progress review. Created comprehensive slides covering company profile, project proposal, methodology, timeline, and expected outcomes.

Presentation Outline Created:

- Company Profile & Microsoft Elevate Learning Modules (Week 1-12 structure)
- Problem Statement addressing Border Defence challenges
- Project Title and Applicability to national security
- Technical Objectives with measurable success metrics
- System Architecture showing data flow and processing layers
- Tech Stack (Python, YOLOv8, Azure, Power BI)
- 12-Week Timeline with detailed weekly milestones
- Deliverables and Success Metrics (>90% accuracy, <20% false positives)

12-Week Project Timeline

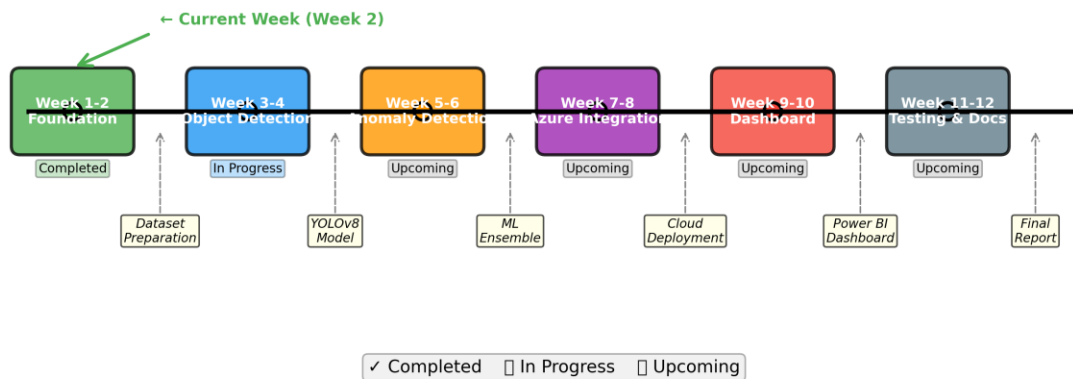


Figure 4: 12-Week Project Implementation Timeline

Plans for next week (Feb 10-14, 2026):

1. Complete Week 3 Microsoft Elevate Modules

- Attend "Power BI Fundamentals" live session
- Learn Power BI Desktop interface and data connections
- Practice creating basic visualizations in Power BI
- Understand Data Modeling concepts

2. Begin Dataset Preparation

- Download sample UCF-Crime dataset (start with 20-30 videos)
- Setup organized folder structure (Data/raw, Data/processed)
- Create dataset documentation (metadata, characteristics)
- Test basic video loading and frame extraction using OpenCV

3. Azure Account Setup and Exploration

- Register for Azure for Students account (free credits)
- Explore Azure portal interface
- Learn about Azure Blob Storage for video storage
- Understand Azure Functions basics for serverless processing

4. OpenCV Learning and Practice

- Complete OpenCV tutorial for video processing
- Practice frame extraction from videos
- Learn basic image preprocessing (resize, normalize)
- Test performance on sample surveillance videos

5. GitHub Repository Setup

- Create GitHub repository: "Border-Surveillance-AI"
- Setup proper folder structure (Data/, Notebooks/, Scripts/, models/)
- Write comprehensive README.md with project overview
- Initialize .gitignore for Python projects

Total Working Hours: 35 hours

Signature of Student: _____

The above entries are correct, and the grading of work done by Trainee is:

Excellent	Very Good	Good	Fair	Below Average	Poor
☐	☐	☐	☐	☐	☐

Signature of External Guide with Company Seal: _____

Date: _____

Signature of Internal Guide: _____

Date: _____